Lecture 10

Dynkin diagrams and the classification

Objectives. *A base of a root system gives an integer matrix and a graph. We compute both for classical algebras. The graph determines the algebra.*

By the end of this lecture you will be able to:

- Compute Cartan matrices and Dynkin diagrams with $a_{ij} = \alpha_j(h_i)$.
- Find the root data and dimensions of types A–D.
- Explain the positivity test behind the list of diagrams.
- Distinguish the classification of root systems from Serre's construction of the algebras.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, root system $R \subset \mathfrak{h}^*$ and base $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$. By Lecture 8, each root α has a *coroot* h_α: the unique element of $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$ with $\alpha(h_\alpha) = 2$. The real span of R carries a positive definite form, invariant under the Weyl group W, with

$$\beta(h_\alpha) = \frac{2(\beta, \alpha)}{(\alpha, \alpha)} \quad \text{for all } \alpha, \beta \in R.$$

Write $h_i = h_{\alpha_i}$.

10.1 Cartan matrices and diagrams

Definition 10.1 (Cartan matrix)**.** The *Cartan matrix* of Δ is the $\ell \times \ell$ matrix A with

$$a_{ij} = \alpha_j(h_i) = \frac{2(\alpha_j, \alpha_i)}{(\alpha_i, \alpha_i)}.$$

The second formula defines it for a base of an abstract root system.

Row i records $[h_i, e_j] = a_{ij} e_j$ for $e_j \in \mathfrak{g}_{\alpha_j}$. Humphreys prints the transpose. Translate his tables before using them.

Proposition 10.2. *Let Δ be a base of a root system R in a Euclidean space.*

(i) $a_{ii} = 2$.

(ii) For $i \neq j$, a_{ij} is an integer ≤ 0, and $a_{ij} = 0$ if and only if $a_{ji} = 0$.

(iii) For $i \neq j$, the number $m_{ij} = a_{ij} a_{ji}$ lies in $\{0, 1, 2, 3\}$.

Proof. Part (i) is the formula with $j = i$. Integrality in (ii) is a root system axiom. Distinct simple roots satisfy $(\alpha_i, \alpha_j) \leq 0$ (Lecture 9; Humphreys §10.1, Lemma). The sign of a_{ij} is the sign of (α_i, α_j), which is symmetric in i and j.

For (iii), let θ be the angle between α_i and α_j. Then

$$a_{ij}a_{ji} = \frac{4(\alpha_i,\alpha_j)^2}{(\alpha_i,\alpha_i)(\alpha_j,\alpha_j)} = 4\cos^2\theta.$$

Simple roots are linearly independent. Cauchy–Schwarz is therefore strict, and the integer $4\cos^2\theta$ satisfies $0 \le 4\cos^2\theta < 4$. ∎

Definition 10.3 (Dynkin diagram)**.** The *Dynkin diagram* of Δ has nodes $1,\dots,\ell$. Distinct nodes i, j are joined by m_{ij} edges. When $m_{ij} \ge 2$, an arrow on these edges points toward the shorter of α_i and α_j.

If $a_{ij} \neq 0$, the formula for A gives

$$\frac{(\alpha_i,\alpha_i)}{(\alpha_j,\alpha_j)} = \frac{a_{ji}}{a_{ij}}.$$

Let $m_{ij} \in \{2,3\}$. Then a_{ij} and a_{ji} are negative integers with prime product m_{ij}, so one of them is -1 and the other is $-m_{ij}$. By the ratio above, the entry $-m_{ij}$ lies in the row of the shorter root. Since $a_{ij} = 0$ for no edge and $a_{ij} = -1$ for a single edge, the diagram determines A.

By Lecture 9, W acts transitively on bases and preserves the form. So another base of R only renumbers A. Cartan subalgebras of $\mathfrak{g}$ are conjugate [C: Humphreys §16.4; Lecture 7]. Hence $\mathfrak{g}$ determines A up to renumbering.

10.2 The classical matrix algebras

For an invertible $N \times N$ matrix J, put

$$\mathfrak{g}_J = \{X \in \mathfrak{gl}_N(\mathbb{C}) : X^{\mathsf{T}}J + JX = 0\}.$$

If $X, Y \in \mathfrak{g}_J$, then $X^{\mathsf{T}}J = -JX$ and $Y^{\mathsf{T}}J = -JY$. The second and third lines below use these identities:

$$\begin{aligned}[X,Y]^{\mathsf{T}}J &= Y^{\mathsf{T}}X^{\mathsf{T}}J - X^{\mathsf{T}}Y^{\mathsf{T}}J \\ &= -Y^{\mathsf{T}}JX + X^{\mathsf{T}}JY \\ &= JYX - JXY \\ &= -J[X,Y].\end{aligned}$$

So $\mathfrak{g}_J$ is a Lie subalgebra.

With $I = I_n$, take the $2n \times 2n$ matrices

$$J_+ = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \qquad J_- = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

For odd size take the $(2n+1) \times (2n+1)$ matrix

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I \\ 0 & I & 0 \end{pmatrix},$$

with rows and columns indexed by $0, 1, \dots, 2n$. Define

$$\mathfrak{so}_{2n} = \mathfrak{g}_{J_+}, \qquad \mathfrak{sp}_{2n} = \mathfrak{g}_{J_-}, \qquad \mathfrak{so}_{2n+1} = \mathfrak{g}_S.$$

We import one fact. The algebras $\mathfrak{sl}_n$ $(n \ge 2)$, $\mathfrak{so}_N$ $(N \ge 3)$ and $\mathfrak{sp}_{2n}$ $(n \ge 1)$ are semisimple [C: Humphreys §19.2; Carter, Chapter 8].

Example 10.4 (The algebra $\mathfrak{sp}_{2n}$)**.** Write $X \in \mathfrak{gl}_{2n}(\mathbb{C})$ in $n \times n$ blocks as

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Block multiplication gives

$$J_- X + X^{\mathsf{T}} J_- = \begin{pmatrix} C - C^{\mathsf{T}} & D + A^{\mathsf{T}} \\ -A - D^{\mathsf{T}} & B^{\mathsf{T}} - B \end{pmatrix}.$$

Thus A is arbitrary, B and C are symmetric, and $D = -A^{\mathsf{T}}$. Consequently

$$\dim \mathfrak{sp}_{2n} = n^2 + 2 \cdot \frac{n(n+1)}{2} = n(2n+1).$$

The diagonal elements $H = \operatorname{diag}(t, -t)$ form an abelian subalgebra $\mathfrak{h}$. Name H by its vector $t = \sum t_i e_i$, and put $\varepsilon_i(H) = t_i$. For every diagonal matrix H and every matrix unit E_{ab},

$$[H, E_{ab}] = (H_{aa} - H_{bb}) E_{ab}.$$

The rule gives these weights.

matrix	indices	weight
$X_{ij} = E_{ij} - E_{n+j,n+i}$	$i \neq j$	$\varepsilon_i - \varepsilon_j$
$E_{i,n+j} + E_{j,n+i}$	$i < j$	$\varepsilon_i + \varepsilon_j$
$E_{n+j,i} + E_{n+i,j}$	$i < j$	$-\varepsilon_i - \varepsilon_j$
$E_{i,n+i},\ E_{n+i,i}$		$\pm 2\varepsilon_i$

The four rows of the table contain

$$n(n-1) + 2 \cdot \frac{n(n-1)}{2} + 2n = 2n^2 = \dim \mathfrak{sp}_{2n} - n$$

matrices, with disjoint supports off the diagonal. So $\mathfrak{sp}_{2n}$ is $\mathfrak{h}$ plus the lines they span, and each nonzero weight occurs once.

Since every listed weight is nonzero, the diagonal, hence toral, algebra $\mathfrak{h}$ is its own centraliser. A toral subalgebra containing $\mathfrak{h}$ is abelian (Lecture 7), so it centralises $\mathfrak{h}$ and equals $\mathfrak{h}$. Thus $\mathfrak{h}$ is a Cartan subalgebra with the listed roots.

For coroots, multiply matrix units:

$$[X_{ij}, X_{ji}] = E_{ii} - E_{jj} - E_{n+i,n+i} + E_{n+j,n+j},$$
$$[E_{n,2n}, E_{2n,n}] = E_{nn} - E_{2n,2n}.$$

The vectors are $e_i - e_j$ and e_n. The roots $\varepsilon_i - \varepsilon_j$ and $2\varepsilon_n$ take the value 2 on them, so these are the coroots.

For B_n and D_n, the Cartan subalgebras, one-dimensional root spaces and coroot data below are imported [C: Humphreys §§1.2 and 19.2]. The next example checks the short coroot of B_n.

Example 10.5 (The short coroot of B_n)**.** For the form S, take

$$e = E_{n,0} - E_{0,2n}, \qquad f = E_{2n,0} - E_{0,n}.$$

Substitution gives $e^{\mathsf{T}}S + Se = f^{\mathsf{T}}S + Sf = 0$, so both lie in $\mathfrak{so}_{2n+1}$. On $H = \operatorname{diag}(0, t, -t)$ the rule gives weights $\pm\varepsilon_n$. Multiplying matrix units gives

$$ef = -E_{nn} - E_{00}, \qquad fe = -E_{2n,2n} - E_{00}.$$

Hence

$$[e, f] = ef - fe = -E_{nn} + E_{2n,2n}.$$

The vector of $[e, f]$ is $-e_n$, and $\varepsilon_n(-e_n) = -1$. Since ε_n takes the value 2 on its coroot, rescale by -2:

$$h_{\varepsilon_n} = -2[e, f] = 2E_{nn} - 2E_{2n,2n}.$$

The vector of h_{ε_n} is $2e_n$, and $\varepsilon_n(2e_n) = 2$. So the short root ε_n has coroot vector $2e_n$.

Take $n \geq 2$ in type A, $n \geq 1$ in B, C, and $n \geq 2$ in D. In type A (Exercise 7.1), $\mathfrak{h}$ consists of the traceless diagonal matrices $\operatorname{diag}(t)$, named by t, with $\varepsilon_i(\operatorname{diag}(t)) = t_i$.

Type	$\mathfrak{g}$	roots	$\lvert R\rvert$	$\dim \mathfrak{g}$
A_{n-1}	$\mathfrak{sl}_n$	$\varepsilon_i - \varepsilon_j$	$n(n-1)$	$n^2 - 1$
B_n	$\mathfrak{so}_{2n+1}$	$\pm\varepsilon_i \pm \varepsilon_j,\ \pm\varepsilon_i$	$2n^2$	$n(2n+1)$
C_n	$\mathfrak{sp}_{2n}$	$\pm\varepsilon_i \pm \varepsilon_j,\ \pm 2\varepsilon_i$	$2n^2$	$n(2n+1)$
D_n	$\mathfrak{so}_{2n}$	$\pm\varepsilon_i \pm \varepsilon_j$	$2n(n-1)$	$n(2n-1)$

Here $i \neq j$ in type A, and $i < j$ in the other two-index expressions. In every type, the simple roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ $(i < \ell)$ have coroot vectors $e_i - e_{i+1}$. The last simple root and its coroot vector are

$$\begin{aligned} A_{n-1}: &\quad \alpha_{n-1} = \varepsilon_{n-1} - \varepsilon_n, &\quad h_{n-1} = e_{n-1} - e_n, \\ B_n: &\quad \alpha_n = \varepsilon_n, &\quad h_n = 2e_n, \\ C_n: &\quad \alpha_n = 2\varepsilon_n, &\quad h_n = e_n, \\ D_n: &\quad \alpha_n = \varepsilon_{n-1} + \varepsilon_n, &\quad h_n = e_{n-1} + e_n. \end{aligned}$$

Proposition 10.6. *In each row, $\alpha_1, \ldots, \alpha_\ell$ form a base.*

Proof. For $i < j$, with empty sums zero,

$$\begin{aligned} \varepsilon_i - \varepsilon_j &= \alpha_i + \cdots + \alpha_{j-1}, \\ B_n:\ \varepsilon_i &= \alpha_i + \cdots + \alpha_n, \\ C_n:\ 2\varepsilon_i &= 2(\alpha_i + \cdots + \alpha_{n-1}) + \alpha_n, \\ D_n:\ \varepsilon_i + \varepsilon_j &= (\alpha_i + \cdots + \alpha_{n-2}) \\ &\quad + (\alpha_j + \cdots + \alpha_{n-1}) + \alpha_n. \end{aligned}$$

In B_n, $\varepsilon_i + \varepsilon_j$ is the sum of two such expressions. In C_n it equals $(\varepsilon_i - \varepsilon_j) + 2\varepsilon_j$. So every root is $\pm$ a nonnegative integer combination of the simple roots. The roots span $\mathfrak{h}^*$, so the $\ell = \dim \mathfrak{h}$ simple roots form a basis. ∎

For $n \geq 2$, the coroot column gives the last rows of A. In C_n,

$$\begin{aligned} a_{n-1,n} &= 2\varepsilon_n(e_{n-1} - e_n) = -2, \\ a_{n,n-1} &= (\varepsilon_{n-1} - \varepsilon_n)(e_n) = -1. \end{aligned}$$

The entry -2 lies in row $n-1$, so α_{n-1} is short. The arrow points toward node $n-1$. In B_n, the coroot vector $2e_n$ gives

$$a_{n-1,n} = \varepsilon_n(e_{n-1} - e_n) = -1,$$
$$a_{n,n-1} = (\varepsilon_{n-1} - \varepsilon_n)(2e_n) = -2.$$

Thus $A(B_n) = A(C_n)^{\mathsf{T}}$, and the arrow points toward node n. In D_n $(n \geq 3)$,

$$a_{n-1,n} = (\varepsilon_{n-1} + \varepsilon_n)(e_{n-1} - e_n) = 0,$$
$$a_{n-2,n} = (\varepsilon_{n-1} + \varepsilon_n)(e_{n-2} - e_{n-1}) = -1.$$

Similarly $a_{n,n-2} = -1$, and $a_{kn} = a_{nk} = 0$ for $k < n-2$. So node n joins node $n-2$ only. For $n \geq 4$ the diagram forks at node $n-2$.

10.3 Positivity and the list of diagrams

Put $u_i = \alpha_i/|\alpha_i|$. For $i \neq j$, Proposition 10.2 gives $4\cos^2\theta_{ij} = m_{ij}$ and $\cos\theta_{ij} \leq 0$. Hence

$$(u_i, u_j) = \cos\theta_{ij} = -\frac{1}{2}\sqrt{m_{ij}}.$$

For $x \in \mathbb{R}^{\ell}$, expand the norm:

$$\Big|\sum_i x_i u_i\Big|^2 = \sum_i x_i^2 + 2\sum_{i<j}(u_i, u_j)\, x_i x_j$$
$$= \sum_i x_i^2 - \sum_{i<j}\sqrt{m_{ij}}\, x_i x_j.$$

For any graph Γ with m_{ij} edges between nodes i and j, the last line defines $q_\Gamma(x)$, the *normalised Gram form* of Γ.

> **Proposition 10.7.** *If Γ is the Dynkin diagram of a root system, then q_Γ is positive definite. So is $q_{\Gamma'}$ for every Γ' obtained by deleting nodes.*

Proof. The u_i are independent, so the norm is positive for $x \neq 0$. Deleting nodes sets their x_i to zero. ∎

For example, let Γ' be a cycle of $k \geq 3$ nodes with single edges. For $x = (1, \dots, 1)$ there are k squares and k edge terms, so

$$q_{\Gamma'}(x) = k - k = 0.$$

So deleting nodes from a Dynkin diagram never leaves a cycle of single edges.

Remark 10.8 (Restrictions behind the list [S]). The same test on suitable subgraphs restricts a connected Dynkin diagram. Such a diagram has no cycles. At most three edges, counted with multiplicity, meet at a node, so a triple edge occurs only in G_2. There is at most one multiple edge or branch node in total, and the arm lengths are bounded. *Omitted step:* these proofs and the final case analysis (Humphreys §11.4; Erdmann–Wildon, Chapter 13).

A root system is *irreducible* if it is not the union of two nonempty orthogonal subsets. Equivalently, its diagram is connected [C: Humphreys §11.3].

Theorem 10.9 (Classification of root systems [C]). *Let R be an irreducible root system of rank ℓ.*

(a) Its Dynkin diagram is one in the table below.

(b) A bijection between a base of R and a base of another root system R' with equal Cartan matrices extends uniquely to an isomorphism $R \to R'$.

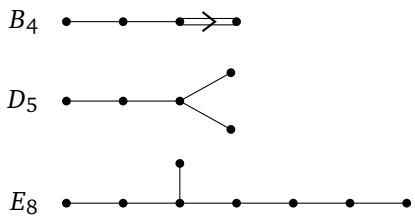
(c) Every diagram in the table belongs to a root system.

Sources: (a) Humphreys §11.4, Theorem; (b) Humphreys §11.1, Proposition; (c) Humphreys §12.1.

The arms of a branch node are counted without that node. The dimensions use $\dim \mathfrak{g} = \ell + |R|$, with exceptional root counts cited [C: Humphreys, Chapter III].

Type	shape	$\dim \mathfrak{g}$
A_ℓ, $\ell \geq 1$	chain of single edges	$\ell(\ell+2)$
B_ℓ, $\ell \geq 2$	chain; double end edge, arrow to the end	$\ell(2\ell+1)$
C_ℓ, $\ell \geq 3$	chain; double end edge, arrow from the end	$\ell(2\ell+1)$
D_ℓ, $\ell \geq 4$	branch node with arms $1, 1, \ell-3$	$\ell(2\ell-1)$
E_6	branch node with arms $2, 2, 1$	78
E_7	branch node with arms $2, 3, 1$	133
E_8	branch node with arms $2, 4, 1$	248
F_4	four-node chain, double middle edge	52
G_2	two nodes, triple edge	14

The figure draws the diagrams B_4, D_5 and E_8.



10.4 Serre relations and reconstruction

By Lecture 8, choose root vectors $e_i \in \mathfrak{g}_{\alpha_i}$ and $f_i \in \mathfrak{g}_{-\alpha_i}$ with $[e_i, f_i] = h_i$.

Proposition 10.10 (Serre relations). *For all i, j,*

$$[h_i, h_j] = 0, \qquad [e_i, f_j] = \delta_{ij} h_i, \qquad [h_i, e_j] = a_{ij} e_j, \qquad [h_i, f_j] = -a_{ij} f_j.$$

For $i \neq j$,

$$(\operatorname{ad} e_i)^{1-a_{ij}}(e_j) = 0, \qquad (\operatorname{ad} f_i)^{1-a_{ij}}(f_j) = 0.$$

Proof. The algebra $\mathfrak{h}$ is abelian. For $i \neq j$, $[e_i, f_j] \in \mathfrak{g}_{\alpha_i - \alpha_j}$. The coefficients of $\alpha_i - \alpha_j$ have mixed signs, so it is neither a root nor zero. Hence $[e_i, f_j] = 0$. The elements e_j and f_j have weights $\pm\alpha_j$, and $\alpha_j(h_i) = a_{ij}$.

For the last two relations, let $i \neq j$. By Lecture 8, the α_i-string through α_j is $\alpha_j - r\alpha_i, \ldots, \alpha_j + q\alpha_i$, with $r - q = \alpha_j(h_i) = a_{ij}$. The coefficients of $\alpha_j - \alpha_i$ have mixed signs, so it is not a root.

Hence $r = 0$ and $q = -a_{ij}$. So $\alpha_j + (1 - a_{ij})\alpha_i$ lies beyond the string. It is neither a root nor zero, and

$$(\operatorname{ad} e_i)^{1-a_{ij}}(e_j) \in \mathfrak{g}_{\alpha_j+(1-a_{ij})\alpha_i} = 0.$$

Negating all roots gives the relation for the f_i. ∎

Theorem 10.11 (Serre [C]). *Let A be the Cartan matrix of a base of a root system R. Let $\mathfrak{g}(A)$ be the Lie algebra generated by e_i, f_i, h_i $(1 \le i \le \ell)$ subject only to the relations of Proposition 10.10.*

(a) $\mathfrak{g}(A)$ is finite-dimensional and semisimple. The span of the h_i is a Cartan subalgebra, with root system isomorphic to R and simple coroots h_i.

(b) If $\mathfrak{g}$ is semisimple with Cartan matrix A, the generators of $\mathfrak{g}(A)$ map to the chosen e_i, f_i, h_i by an isomorphism $\mathfrak{g}(A) \to \mathfrak{g}$.

Source: Humphreys §§14.2 and 18.3–18.4. His printed matrix is transposed.

Root-system classification uses Euclidean geometry. Serre's theorem supplies existence and uniqueness of the algebra with the prescribed Cartan matrix.

By a cited theorem [C: Humphreys §14.1], a semisimple algebra is simple exactly when its root system is irreducible. Hence each complex simple Lie algebra is isomorphic to the algebra of exactly one type in the table. For types A–D it is a matrix algebra of § 10.2.

10.5 Low rank

The low-rank diagrams give

$$\mathfrak{so}_5 \cong \mathfrak{sp}_4, \qquad \mathfrak{so}_6 \cong \mathfrak{sl}_4.$$

For $B_2 = C_2$, exchange the two nodes. For $D_3 = A_3$, order the nodes as $\alpha_2, \alpha_1, \alpha_3$. Serre's theorem gives the algebra isomorphisms. Exercises 10.6 and 10.7 work through the root identifications.

For D_2, the two simple roots are orthogonal, so the diagram consists of two separate nodes. The corresponding root triples commute and span the algebra:

$$\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2.$$

Exercise 10.7 gives the bracket argument.

Within the ranges of the classical table, the remaining coincidence is $B_1 = C_1 = A_1$, that is, $\mathfrak{so}_3 \cong \mathfrak{sl}_2 = \mathfrak{sp}_2$.

Example 10.12 (G_2). In this example, $\varepsilon_1, \varepsilon_2, \varepsilon_3$ denote the standard basis vectors of $\mathbb{R}^3$. In the plane

$$\{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$$

with the dot product, take the short roots $\pm(\varepsilon_i - \varepsilon_j)$ and the long roots $\pm(2\varepsilon_i - \varepsilon_j - \varepsilon_k)$, $\{i, j, k\} = \{1, 2, 3\}$. By Humphreys §12.1 [C], these 12 vectors form a root system with base

$$\alpha_1 = \varepsilon_1 - \varepsilon_2, \qquad \alpha_2 = -2\varepsilon_1 + \varepsilon_2 + \varepsilon_3.$$

The dot products are $(\alpha_1, \alpha_1) = 2$, $(\alpha_2, \alpha_2) = 6$ and $(\alpha_1, \alpha_2) = -3$. Hence

$$a_{12} = \frac{2(-3)}{2} = -3, \qquad a_{21} = \frac{2(-3)}{6} = -1,$$

$$A(G_2) = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

Theorem 10.11 gives an algebra of dimension $2 + 12 = 14$. It is simple because its diagram is connected.

The types F_4, E_6, E_7, E_8 are identified by their diagrams in the table. They exist by Theorems 10.9(c) and 10.11(a). We do not construct them.

Self-checks

(1) Identify the type of the Cartan matrix

$$\begin{pmatrix} 2 & -1 & 0 \\ -2 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

(2) In B_2 with $a_{12} = -1$ and $a_{21} = -2$, which powers of $\operatorname{ad} e_1$ and $\operatorname{ad} e_2$ appear in the Serre relations?

(3) Are $\mathfrak{so}_7$ and $\mathfrak{sp}_6$ isomorphic?

Answers. (1) Since $a_{21} = -2$, α_2 is short, and α_3 has the same length. So α_1 is the only long simple root, and it is an end node. This is type C_3, numbered in reverse. (2) $(\operatorname{ad} e_1)^2(e_2) = 0$ and $(\operatorname{ad} e_2)^3(e_1) = 0$. (3) No. Both have dimension 21. Isomorphic algebras have the same Cartan matrix up to renumbering (§ 10.1). But B_3 has one short simple root and C_3 has two.

Summary

A base gives a Cartan matrix, and the diagram records the same data. Positivity of q_Γ is the test behind the classification, which restricts connected diagrams to types A–G. The classical algebras realise A–D. Serre's theorem shows that the Cartan matrix determines the algebra. With the classification theorem, it shows that every type occurs.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 11–12, 14 and 18–19. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapters 12–14. Carter, *Lie Algebras of Finite and Affine Type*, Chapter 8.